



Donna Barbar Polytechnic College

Management System (DBPCMS)

Comprehensive User Manual

Version 1.0 • Generated 8/9/2026

Roles covered: Administrator, HR Officer, Registrar, Department Head, Instructor, Dean, Employee

Contents

1. Introduction & Overview
2. Getting Started
3. User Roles & What Each Can Do
4. System Administrator Guide
5. Human Resource Officer Guide
6. Registrar Guide
7. Department Head Guide
8. Instructor Guide
9. Dean Guide
10. Employee (Self-Service) Guide
11. Grading: How Marks Become Grades
12. Transcripts & Document Verification
13. Reports & Exports
14. Security, Privacy & Data Protection
15. Backups & Recovery
16. Troubleshooting & FAQ
17. Glossary

1. Introduction & Overview

The Donna Barbar Polytechnic College Management System (DBPCMS) is a secure, web-based platform that digitizes the college's employee management and student grading. It replaces paper files with a centralized, role-based system used by administrators, HR officers, registrars, deans, department heads, instructors, and employees.

What the system does

- Employee Management: staff records, education, qualifications, employment history, emergency contacts, documents, and photos.
- Student Academic Management: students, courses, sections, instructor assignment, and enrollment.
- Configurable Grading: grade components and weights, grading scales, and GPA rules — editable without changing code.
- Grade Workflow: instructors enter marks; grades are submitted, approved, and published through a controlled, audited chain.
- Transcripts & Reports: official transcripts with GPA and QR verification, plus HR and academic analytics (PDF/Excel/CSV).
- Security & Auditing: role-based access, secure passwords, and a permanent audit trail.

Key principles

- Role-based: you only see and do what your job requires.
- Configurable: business rules are set in the interface, not hard-coded.
- Traceable: important actions are recorded in an audit log.
- Bilingual-ready: English and Afaan Oromo interface.

2. Getting Started

Logging in

1. Open the system's web address in a browser (Chrome or Edge recommended).
2. Enter your Email and Password. Use the eye icon to reveal your password if needed.
3. Click Sign in.
4. On first login you must set a new password.

NOTE

Your account is created by an administrator. If you cannot log in, ask your System Administrator to reset your password.

Passwords

- At least 12 characters, with uppercase, lowercase, and a number.
- Changing your password signs you out of all sessions; log in again with the new one.

The interface

- Left sidebar: navigation — shows only what your role can access.
- Top bar: global search, language switch, light/dark mode, notifications, and your account menu.
- Main area: lists, forms, dashboards, and reports.

Common controls

- Search boxes filter lists.
- Tables page with Prev/Next.
- Create/Edit open dialog forms with validation.
- Destructive actions ask for confirmation.
- Brief toast messages confirm success or show errors.

3. User Roles & What Each Can Do

DBPCMS uses Role-Based Access Control (RBAC). Each user has one or more roles; a role bundles permissions.

Role **What they can do**

System Administrator	Full control: users, roles, departments, programs, academic calendar, settings, audit logs, backups.
HR Officer	Register/manage employees, upload documents, print profiles, generate HR reports.
Registrar	Manage students and semesters; publish grades; generate transcripts; unlock grades; academic reports.
Department Head	Manage instructors and courses/sections; approve grades for their department; departmental analytics.
Instructor	View assigned sections; enter/edit (pre-submission)/submit grades; view class statistics.
Dean	Institutional oversight: reports, analytics, employee and academic statistics.
Employee	View own profile and documents; update limited personal information.

4. System Administrator Guide

The administrator sets up and maintains the system. Recommended first-time setup order:

1. Create Departments (optionally assign a Head).
2. Create Programs under each department.
3. Create the Academic Year and mark it current.
4. Create Semesters within the year.
5. Configure Grading Setup: components + weights, and the grading scale.
6. Create staff user accounts and assign roles.
7. Review System Settings.

Managing users & roles

- Administration !' Users & Roles !' New user: name, email, temporary password, and role(s).
- The user changes the temporary password at first login.
- Key icon resets a password; pencil edits; trash removes.
- Guards prevent deactivating/deleting your own account or removing the last administrator.

System settings

- Retirement age (Retirement List report).
- Contract expiry window in days (Contract Expiry report).
- Institution name (shown on printed documents).
- Student/Employee ID prefixes (e.g. DBPC-STU / DBPC-EMP).

Audit log

Administration !' Audit Log: a read-only history of critical actions (who, what, when). It cannot be edited or deleted.

5. Human Resource Officer Guide

Registering an employee

1. Employees !' Register employee.
2. Fill personal details (name, gender, DOB, National ID, phone, email) and employment details (department, position, type, status, date of employment).
3. Save. A unique Employee ID is generated (e.g. DBPC-EMP-2026-00001).

Completing the profile

- Open an employee (View) to add Education, Qualifications, Employment History, and Emergency Contacts.
- Documents tab: upload degrees, CV, contracts, ID scans (PDF, DOCX, XLSX, JPG, PNG up to 10 MB); images are compressed.
- Upload a portrait photo on the profile header (appears on the printed profile).

Printing an official profile

1. Open the employee !' Print.
2. An A4 page opens with details, photo, and a QR verification code.
3. Use the browser's Print / Save as PDF.

HR reports

Reports !' HR Reports: Employee Directory, Department Employees, Qualification Summary, New Employees, Contract Expiry, Retirement List, Staff Distribution. Export to PDF/Excel/CSV.

6. Registrar Guide

Managing students

1. Students !' Register student: details + department and program.
2. Save. A unique Student ID is generated (e.g. DBPC-STU-2026-00001).
3. Optionally upload a student photo (used on the transcript) via the edit form.

Enrolling students

- Academic !' Sections !' Manage !' Students: enroll students into a section.
- A student must be enrolled to receive grades for that course.

Publishing grades

1. After a Department Head approves a section, open Grade Entry and select it.
2. Click Publish: results are computed, a snapshot is frozen, and grades are locked.
3. For post-publish corrections, use Unlock (registrar/admin), then re-run the workflow.

Transcripts

Transcripts !' search or browse (by department/program or by section) !' select a student !' view GPA !' Print transcript (A4 with QR and photo).

7. Department Head Guide

- Manage courses and sections for your department (Academic).
- Assign instructors to sections (Academic !' Sections !' Manage !' Instructors).
- Approve grades when an instructor submits a section, or Return for correction with a reason.
- You may enter grades for your department's sections if needed.
- You can only approve grades for your own department (enforced).

NOTE

An administrator must set you as the Head of your department (Departments !' edit !' Head) for department-scoped actions.

8. Instructor Guide

Entering grades

1. Grade Entry !' choose the semester !' open your section (you only see sections assigned to you).
2. Type each student's raw mark per component (e.g. out of 25 for the Mid Exam). You cannot exceed a component's maximum.
3. Letter grade and percentage compute live; work autosaves after a short pause.
4. Click Submit to send grades for approval.

After submitting

- While submitted or approved, the sheet is locked to you.
- If Returned for correction, read the reason, edit, and re-submit.
- Once the Registrar publishes, grades are final and locked.

9. Dean Guide

- Dashboard: institution-wide counts and recent activity.
- Reports !' Academic Analytics: Pass Rate, Top Students, Average GPA by Department, Failure Analysis.
- Reports !' HR Reports for staffing insights.
- Export any report to PDF/Excel/CSV.

10. Employee (Self-Service) Guide

Employees with a login can view their own profile and documents and update limited personal information. Broader self-service is planned for a future release.

11. Grading: How Marks Become Grades

The pieces (all configurable)

- Components: parts of assessment (Quiz, Assignment, Mid, Final), each with a maximum score and a weight percentage. Active weights should total 100%.
- Grading scale: percentage bands mapped to letters and grade points (e.g. 90-100 = A+ = 4.0), plus a pass mark and rounding rule.
- GPA: the credit-weighted average of grade points.

The calculation

1. Each component's raw mark becomes a fraction of its maximum.
2. Fractions combine by weight into an overall percentage.
3. The percentage is rounded per the configured rule.
4. The rounded percentage is matched to a grade band -> letter, grade point, pass/fail.
5. $GPA = \frac{\text{sum}(\text{grade point} \times \text{credit hours})}{\text{total credit hours}}$

NOTE

On publish, the exact scale, weights, and rules are frozen with each result. Later configuration changes never alter already-published grades or transcripts.

12. Transcripts & Document Verification

- Transcripts and printed profiles include a QR code and a short verification code.
- Anyone can confirm authenticity at the public /verify page: scan the QR or type the code.
- The check shows only a safe summary (document type, subject name, issue date).

13. Reports & Exports

All reports can be viewed on screen and exported using the PDF, Excel, or CSV buttons.

HR Reports

- Employee Directory
- Department Employees
- Qualification Summary
- New Employees
- Contract Expiry
- Retirement List
- Staff Distribution

Academic Analytics

- Pass Rate by Department
- Top Students by GPA
- Average GPA by Department
- Failure Analysis by Course

14. Security, Privacy & Data Protection

- Passwords are stored with strong one-way hashing; never visible to anyone, including administrators.
- Access is controlled by role and permission, re-checked on every request.
- Repeated failed logins temporarily lock an account.
- Sensitive fields (e.g. National ID) are permission-gated and access is logged.
- All critical changes are recorded in the audit log.
- Use the system over HTTPS in production; never share your password.

15. Backups & Recovery

- Database and uploaded files are backed up by scripts/backup.sh.
- Backups are timestamped, compressed, and rotated (default 14 days).
- Schedule nightly backups with cron (see docs/backup-runbook.md).
- Restore with scripts/restore.sh <backup-file>; test restores periodically.
- Keep an off-machine copy of backups on the server.

16. Troubleshooting & FAQ

I can't log in

Check email/password and Caps Lock. If it persists, ask an administrator to reset your password.

A menu item I expected is missing

Menus are filtered by role. Ask an administrator to adjust your role, then log out and back in.

A page says 'Something went wrong'

Click Retry. If it continues, note what you did and the time, and report it to the administrator.

I can't edit grades

Grades lock once submitted, approved, or published. A returned sheet can be edited by the instructor; a published sheet must be unlocked by the Registrar.

A new menu didn't appear after an update

Log out and back in to refresh your permissions.

17. Glossary

Term	Meaning
------	---------

Role	A named bundle of permissions assigned to a user.
Permission	A single ability, such as 'approve grades'.
Section	A specific offering of a course in a semester.
Enrollment	A student registered in a section; grades attach here.
Component	A graded part of a course with a max score and weight.
Grading scale	Mapping of percentages to letters and grade points.
GPA	Grade Point Average — credit-weighted average of grade points.
Publish	Finalize and lock a section's grades; freezes a snapshot.
Audit log	A permanent record of who did what and when.
Transcript	An official record of a student's courses, grades, and GPA.